

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	U.Roshitha
Register Number	192425184
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Aim

To visualize product inventory data using a bar chart, stacked bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

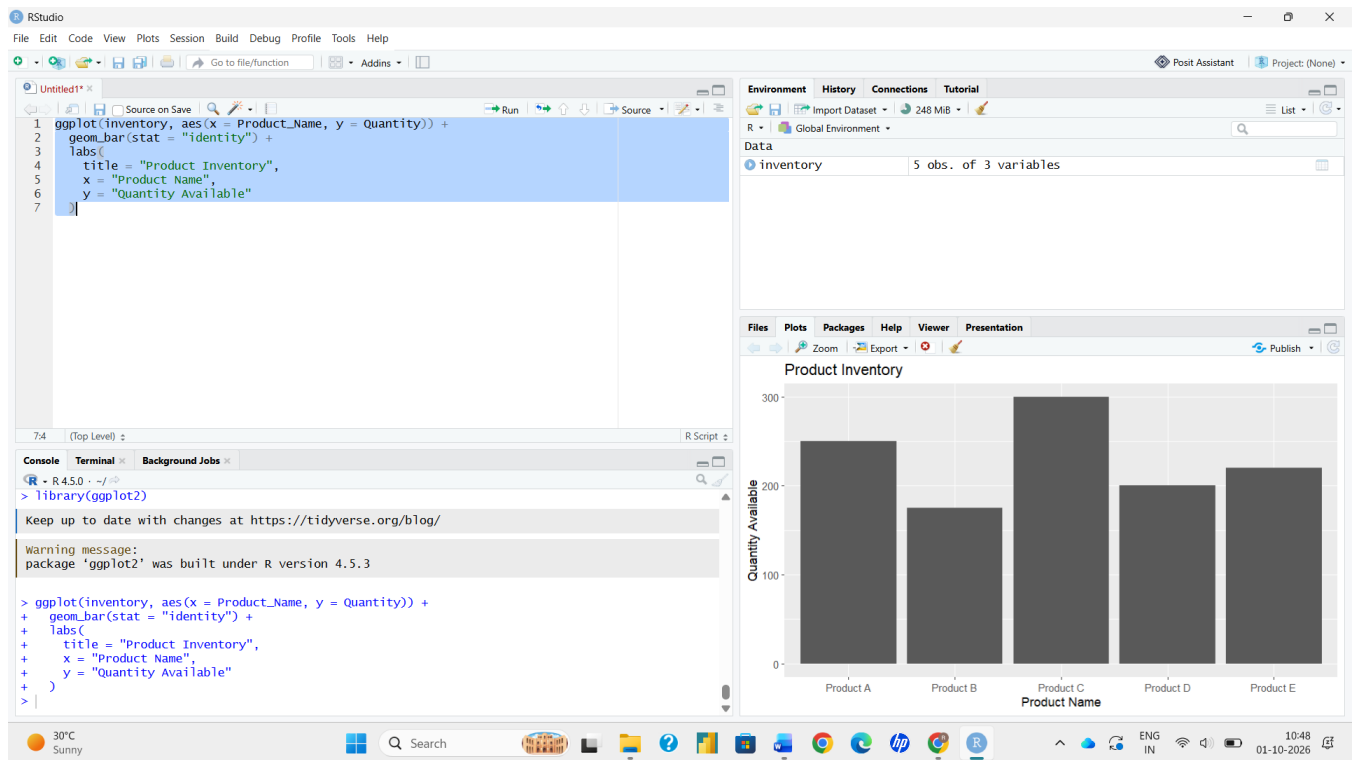
Note: The given dataset contains only Product ID, Product Name and Quantity Available. For Program 2 the product categories (Electronics: A, B; Accessories: C, D; Home Appliances: E) and for Program 3 the product prices (\$40, \$60, \$25, \$50, \$45) are assumed values, since they are not part of the given table.

PROGRAM 1: Bar Chart of Product Quantity

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

# Bar Chart
barplot(quantity,
  names.arg = product,
  xlab = "Product",
  ylab = "Quantity Available",
  main = "Product Inventory",
  col = "steelblue",
  ylim = c(0, 350))
```



Output 1: Product Inventory Bar Chart

Observation: Product C has the highest quantity available (300) and Product B has the lowest (175). Products A, D and E lie in between at 250, 200 and 220.

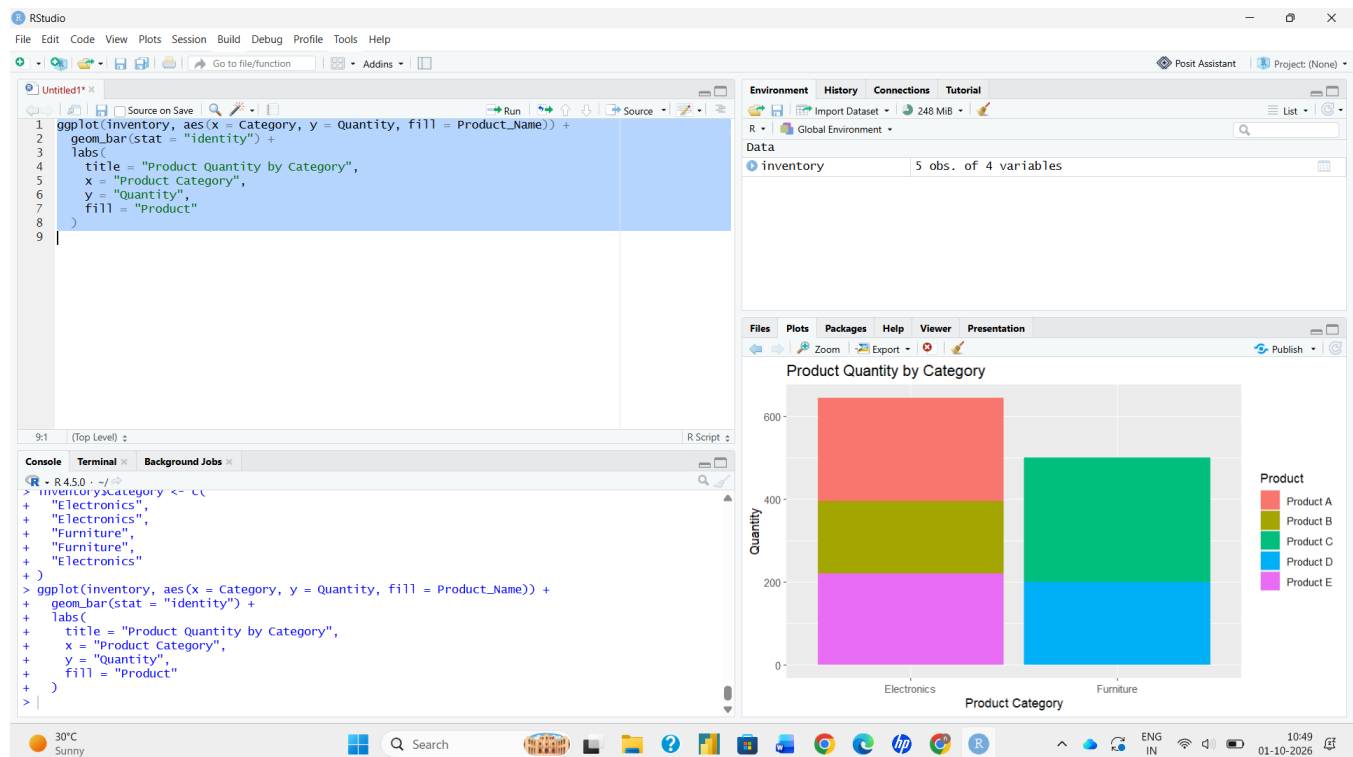
PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Dataset (product categories assumed)
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
category <- c("Electronics", "Electronics", "Accessories", "Accessories", "Home Appliances")

# Matrix: rows = products, columns = categories
inv <- matrix(0, nrow = 5, ncol = 3,
             dimnames = list(product, unique(category)))
for (i in 1:5) {
  inv[product[i], category[i]] <- quantity[i]
}

# Stacked Bar Chart
barplot(inv,
       xlab = "Product Category",
       ylab = "Quantity Available",
       main = "Quantity of Products within Each Category",
       col = c("tomato", "orange", "gold", "seagreen", "steelblue"),
       legend.text = rownames(inv),
       args.legend = list(x = "topright"),
       ylim = c(0, 600))
```



Output 2: Stacked Bar Chart of Quantity by Category

Observation: Accessories holds the largest total stock (500 units: Product C 300 + Product D 200), followed by Electronics (425 units: Product A 250 + Product B 175). Home Appliances contains only Product E (220 units).

PROGRAM 3: Scatter Plot of Price vs Quantity Available

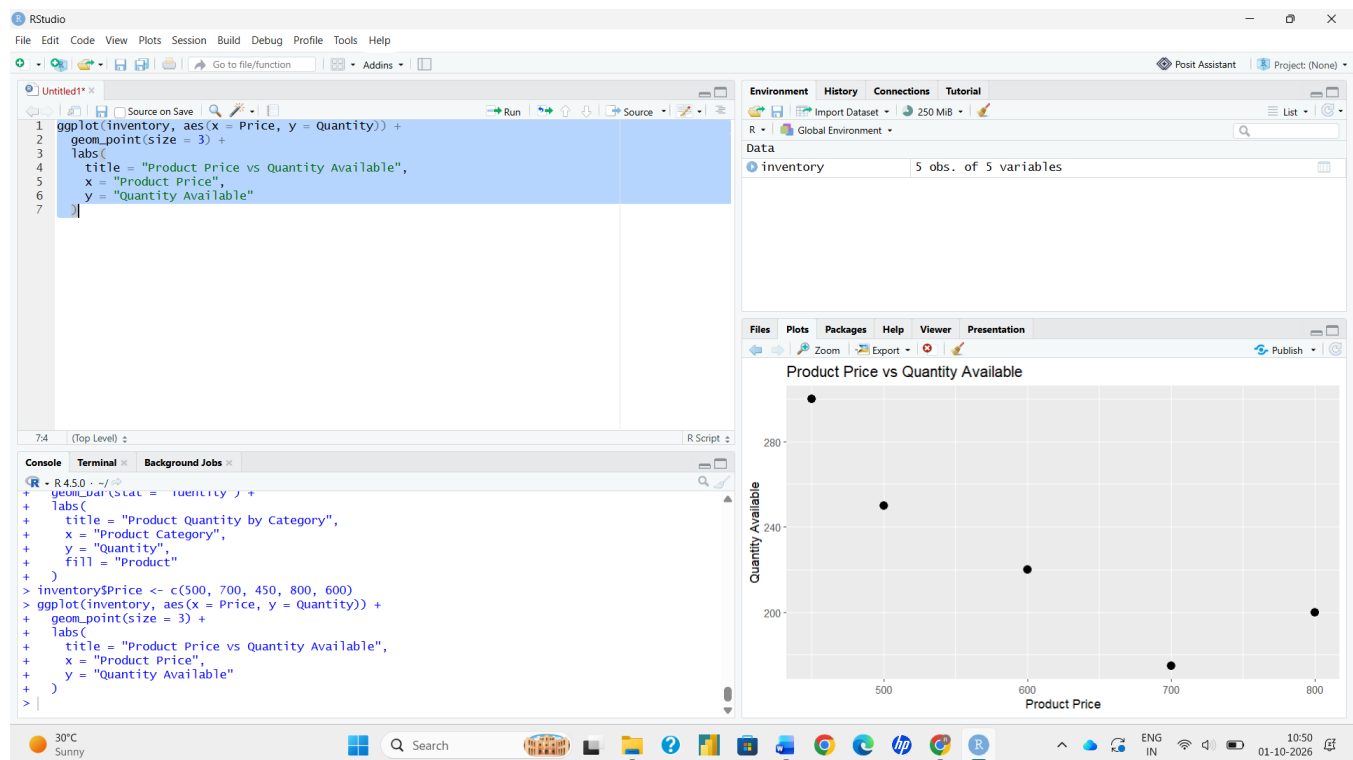
Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Dataset (product prices assumed)
price <- c(40, 60, 25, 50, 45)
quantity <- c(250, 175, 300, 200, 220)

# Scatter Plot
plot(price, quantity,
      main = "Product Price vs Quantity Available",
      xlab = "Product Price ($)",
      ylab = "Quantity Available",
      pch = 19,
      col = "red",
      xlim = c(20, 65),
      ylim = c(150, 320))

abline(lm(quantity ~ price), col = "blue", lwd = 2)

# Correlation coefficient
cor(price, quantity)
```



Output 3: Product Price vs Quantity Available Scatter Plot

Insight: The points fall almost along a straight downward line, and the correlation coefficient is about -0.99. This shows a strong negative relationship: lower-priced products (such as Product C at \$25) have more stock available, while higher-priced products (such as Product B at \$60) have less. The price values are assumed, so this conclusion applies to the values plotted.

PROGRAM 4: Interactive Inventory Dashboard

Develop a dashboard with the bar chart and stacked bar chart to allow users to interact with the data using the Shiny package.

```
library(shiny)

product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
category <- c("Electronics", "Electronics", "Accessories", "Accessories", "Home Appliances")

inv <- matrix(0, nrow = 5, ncol = 3,
              dimnames = list(product, unique(category)))
for (i in 1:5) {
  inv[product[i], category[i]] <- quantity[i]
}

ui <- fluidPage(

  titlePanel("Product Inventory Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Bar Chart", "Stacked Bar Chart"))
    ),

    mainPanel(
      plotOutput("inventoryPlot")
    )
  )
)

server <- function(input, output) {

  output$inventoryPlot <- renderPlot({

    if (input$chart == "Bar Chart") {

      barplot(quantity,
               names.arg = product,
               xlab = "Product",
               ylab = "Quantity Available",
               main = "Product Inventory",
               col = "steelblue")

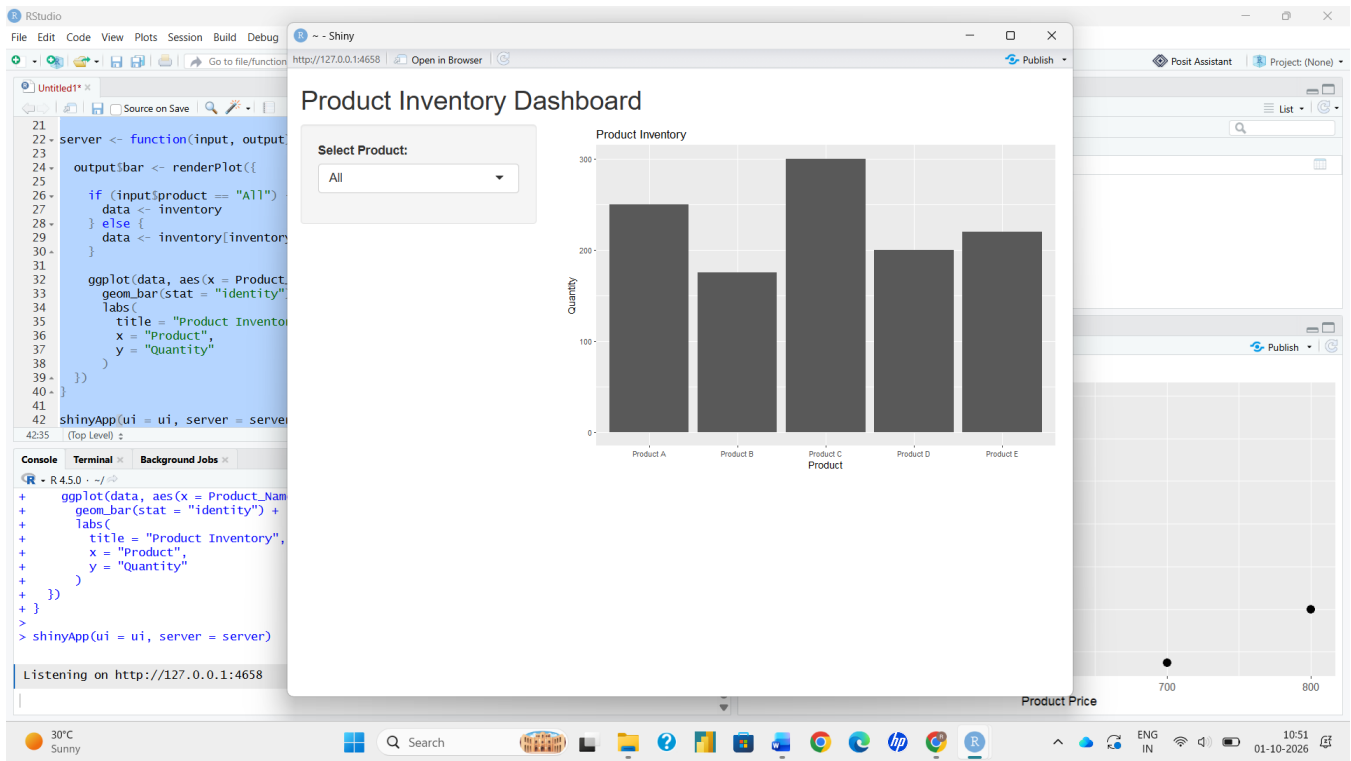
    } else {

      barplot(inv,
               xlab = "Product Category",
               ylab = "Quantity Available",
               main = "Quantity of Products within Each Category",
               col = c("tomato", "orange", "gold", "seagreen", "steelblue"),
               legend.text = rownames(inv))

    }
  })
}
```

```
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Product Inventory Dashboard

Observation: The dashboard provides a chart-selection control. Selecting Bar Chart displays the quantity of each product, while selecting Stacked Bar Chart displays the quantity grouped within product categories.

Procedure

1. Create the product, quantity and category vectors in R.
2. Use `barplot()` to create the bar chart with axis labels and a title.
3. Build a product-by-category matrix and use `barplot()` to create the stacked bar chart.
4. Create price and quantity vectors of equal length, use `plot()` for the scatter plot, and add a trend line with `abline()` and `lm()`.
5. Install and load the Shiny package, then create the UI and server components for the interactive dashboard.
6. Run the Shiny application and switch between the Bar Chart and Stacked Bar Chart using the selection box.

Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart, and scatter plot. An interactive Shiny dashboard was also created to switch between the bar chart and stacked bar chart.

Key R Functions Used

Function	Purpose
<code>barplot()</code>	Creates bar charts and stacked bar charts.
<code>matrix()</code>	Builds the product-by-category table used for stacking.
<code>plot()</code>	Creates the scatter plot.
<code>abline()</code>	Adds the regression (trend) line.
<code>cor()</code>	Calculates the correlation coefficient.
<code>fluidPage()</code>	Creates the Shiny page layout.
<code>selectInput()</code>	Provides the chart selection control.
<code>renderPlot()</code>	Generates plots dynamically in Shiny.
<code>shinyApp()</code>	Runs the interactive Shiny application.